



Change Processes and Procedures in Service Oriented Virtual Organizations and Collaborative Network

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ABSTRACT

Business processes and services of Service Oriented Virtual Organizations are subject to change to meet the internal and external requirements of the competitive and rapidly changing environment they operate in. Efficient and practical change management solutions are needed to enable partners to gain insight on the procedures and the processes which can be used to facilitate the process of change. This paper presents a procedural change management framework to facilitate the process of change allowing the participating partners in a Virtual Organization to initiate, assess, collaborate, authorize, implement, evaluate and control changes in the SOVO. The solution consists of a multi-layered procedural framework, including the six layers of change processes, change actors, change control requirements and related management interfaces. It is derived from Information Technology Infrastructure Library (ITIL V3), Engineering Change Management (ECM) and European Collaborative Networked Organizations Leadership initiative (ECOLEAD) best practices and recommendations and customized to fit SOVO requirements and challenges. We present the implementation of six layers of changes processes using the IBM Business Process Manager (BPM) and the implementation architecture of the change management console to facilitate the process of change in SOVO.

Categories and Subject Descriptors

H.5.3 [Information Systems]: General - *Collaborative computing*

General Terms

Management

Keywords

Change Management, Change Collaboration, Change Process and Procedures, Virtual Organizations.

1. INTRODUCTION

The evolution of web technologies and services has transformed the internet from a mere repository of information and data into a useful and inevitable platform for service provisioning and sharing. It allows geographically distributed businesses to share information, knowledge and resources to increase collaboration to form collaborative networked organizations where they share risk, skills, resources, and core competencies to add more competitive advantages to their business [1]. The consequences of this, is the emergence of the virtual organization (VO), which is a dynamic, temporal consortium of autonomous legally independent organizations that collaborate with each other to achieve certain business objectives [2].

Since today's business environments are very competitive and rapidly changing; organizations are increasingly changing and improving their internal business processes, services, information systems, and partnerships with other companies to add more value to their businesses [2]. In the same way, VOs need to evolve to meet the market and customer demands by changing their own business processes and services. Managing changes in such complicated business environment with distributed partners manually is not practical. A more practical and efficient way is to have automated or semi-automated process of change, which engages all the VO partners in the process of change. This will allow quicker response time for change, less effort from the VO change manager to track, maintain and coordinate for change and more user transparent ways of change management. Thus, change management processes and procedures are needed to facilitate and manage the process of change in a collaborative way between different partners in this collaborative environment.

In this paper, we present a multi-layered procedural change management framework for SOVO which consist of six layer processes of change, change management actors, change control requirements and related management interfaces. The procedures of change increase flexibility and agility in changing shared-services and processes between partners by engaging all of them in the process of change. The rest of the paper is organized as follows: In Section 2, we introduce our research motivation. In Section 3, we discuss related works by referring to the concept of virtual organization (VO), service oriented architecture (SOA) and SOVO change management. In section 4, we present the proposed procedural change management framework. The implementation of the change processes in the SOVO is described in Section 5. Finally, we conclude the paper in Section 6.

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2. MOTIVATION

VOs are continuously changing because of the dynamic environments in which they operate [3]. The idea of combining the SOA with the VO presents many advantages in today's business environments, but it also introduces new research challenges. The various changes in the SOVO environment are expected to be organized and managed in a way that allows participated partners to evolve and change their own business processes and services accordingly. Therefore, a framework that identifies the impact and triggers of change, in addition to the change processes and procedures for handling all levels of changes plays a central role in the successful deployment, operation and evolution of the SOVO. A structure change management framework which identifies the triggers and levels of change has been presented in [4]. In this paper, we present the procedural change management framework which identifies the processes and procedures of change.

Consider a VOWireless where a Cell Phone service provider, a Wireless Network service provider and a Shipping service provider collaborate with each other to provide a final product which is a cell phone with an activated wireless plan shipped to the customer location. All the service providers share their business processes and services, and collaborate together to form a Virtual Organization, called VOWireless. VOWireless can expect changes in its business processes and services from different sources, where changes could be triggered for different reasons. For instance, if the performance level of shipping service provider does not meet the agreed level of service, then VOWireless needs to react to this trigger of change by following predefined processes and procedures which facilitate the process of change. Since VOs faces different kinds of changes in order to meet the markets and customers' demands, a VO such as VOWireless should employ a change management mechanism which identifies the levels, triggers of change and provide step by step processes and procedures to facilitates the process of change by engaging all partners in the change process in a collaborative and efficient way. The proposed change management processes and procedures in SOVO will facilitate change while also reducing user effort, which eventually leads to faster VO change adoption.

3. RELATED WORK

3.1 Virtual Organizations

VOs are a very important manifestation within Collaborative Networked [5]. The VO participating partners share risks, costs, and benefits, coordinate the sharing of skills, resources and competencies to achieve their operations and use computer networks for interactions [2], [5]. The collaboration between different organizations to form a VO involves mutual engagement of autonomous partners to solve problems that benefits every one of them. This requires mutual trust, commitment, governance and management between participants to ensure the survival of the VO [2], [6]. VO management is "the organization, allocation and co-ordination of resources, their activities and inter-organizational dependencies to achieve the objectives of the VO within the required time, costs and quality frame" [7]. The challenges of VO management include distribution, dependencies and coordination challenges, change management challenges, performance management challenges, knowledge and information management

challenges, risk management challenges, and decision making challenges [8], [9].

3.2 Service Oriented Architecture

Service Oriented Architecture (SOA) is a combination of services that can be exposed to customers and partners where business capabilities and logic can be recombined repeatedly to serve business needs on the long run [10]. SOA is an architectural style where systems consist of service users and service providers. It is an architecture that consists of different tools that support the faster implementation of services [11]. Building business applications using SOA will increase the agility and the flexibility of business processes. Furthermore, it will increase the interoperability between various services and their reusability where different SOA characteristics can provide more advantages to organizations who adopt this architecture [12]. The SOA provides a proper way to create dynamic and collaborative applications that are built for change, because it is able to adapt to the dynamic business environment [10].

3.3 Change Management in SOVO

We are focusing on the technical aspect of change management, which is "a process whereby changes to a service are formally introduced, approved before deployment into the next testing or production stage"[10]. We focus on changes to a service through its whole life-cycle and ensures of efficient handling of all changes through the use of standardized methods and procedures based on industry best practices [10]. The goal is to support the process of change, enable the traceability of changes, minimize the impact of change and gradually improve the day to day operations [13].

Changes in SOVO are triggered for different reasons and in different levels during the operation phase of the VO which may cause a problem to maintain the VO's operations and activities [14]. Different unexpected events (Triggers of change) may lead to changes in the VO which require decision making, problem solving, discussion and collaboration between independent and autonomous partners [5]. A specific change to a service may force a series of changes due to the dependency between services and business processes [15]. The fact that SOVO three modeled layers; which include the service providers' layer, the collaborative process layer and the value network layer, are dependent to and connected with each other makes the process of change in SOVO more complex especially in a high level of change [4]. A structural change management framework has been presented in Obidallah & Raahemi (2014) where the levels and triggers of change in the SOVO have been identified and studied.

4. THE PROPOSED PROCEDURAL CHANGE MANAGEMENT FRAMEWORK

The proposed SOVO procedural change management framework is derived from ITIL V3, ECM and ECOLEAD best practices and recommendations. Figure 1 shows the proposed change management procedural framework and its different components. The following discuss the details components of the framework.

4.1 The six layers of change processes

The six layers of change processes provide a sequence of steps and activities that the VO and its participated organizations can follow in initiating changes to their business processes or services.

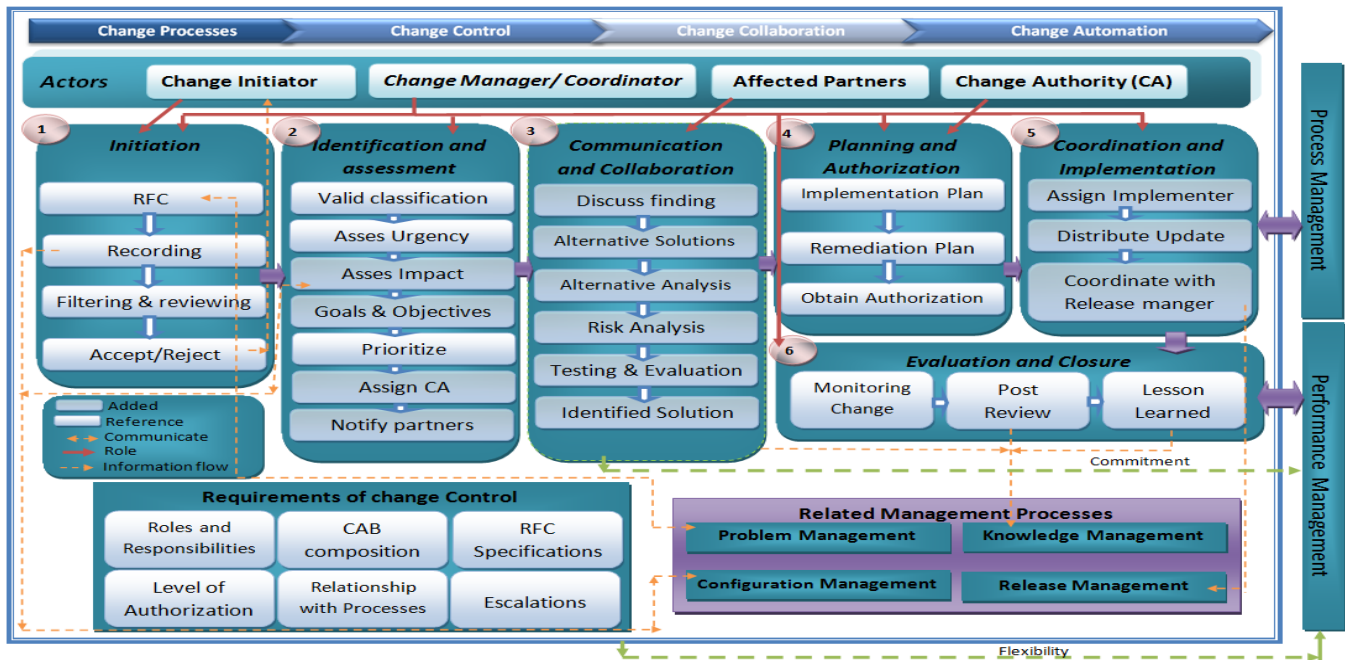


Figure 1: The proposed change management procedural framework

4.1.1 Change Initiation layer

Based on the different triggers of change, changes can be initiated through pre-communication between the VO coordinator and the change initiator which leads to Request for Change (RFC). The RFC is usually initiated because of a problem that requires a change in one of the levels in VO. The request should be recorded, reviewed and filtered by the VO coordinator or the change manager. If any information is missing, it will be returned back to the initiator. The request will be accepted or maybe rejected if further information is needed or a proposal is required.

4.1.2 Identification and Assessment layer

Figure 2 illustrates some of the procedures in the identification and assessment layer where there are different activities to identify:

- Change classification where any change can be classified based on the level of change where the change has started, this include value network level changes, collaborative processes level changes and service level changes.
- Change urgency which is the extent to which a change can bear delay. This can be determined by the change initiator and the change manager. It can be high, medium or low urgency.
- Change impact which is the effect of change on VO. This can be determined based on the change classification. High impact changes are value network changes which affect the VO configuration and topology. Medium impact changes are collaborative process changes which affect the VO choreography. Low impact changes are service level changes which affect the level of the service.
- Change priority provides an order in which a change request in SOVO is considered for authorization. It can be decided based on the impact and the urgency of change.
- Assigning change authority where changes are authorized based on assessment information in the change request details. The change authority can be the VO change advisory board (CAB) members or the change manager based on the level of change.

The change manager can grant the authorization of a change to the change initiator in low level changes.

- Notification for change collaboration and decision making between VO affected partners to find a solution of the problem if needed.

Priority			Change Authority		Impact	
Impact	Urgency	Priority				
High	High	Critical	CAB		High	VNL
High	Medium	High				
High	Low	Moderate				
Medium	High	High	Change manager with CAB confirmation		Medium	CPL
Medium	Medium	Moderate				
Medium	Low	Low	Primary: Change manager May granted to the Change initiator		Low	SL
Low	High	Moderate				
Low	Medium	Low				
Low	Low	Planning				

Figure 2: Identification and assessment layer procedures

4.1.3 Communication and Collaboration layer

All affected partners collaborate to find a solution of the proposed problem when notifications for collaboration are triggered. The VO affected partners collaborate with each other by discussing the change request, providing solutions, risk analysis, ways to test and evaluate their solutions before implementing it. Partners can discuss the problem in a collaborative environment using three kinds of collaboration ways before providing their final solutions. One solution is selected by voting. Only the eligible voting members are able to engage in the voting process which includes the VO change manager, the VO CAB members, the VO coordinator, and an invited expert.

4.1.4 Planning and Authorization layer

Change implantation and remediation plans are developed, and the authorization of a change is requested from the change authority (CA). The change window; which is the agreed time when the change will be implemented with minimal impact on

VO operations, is selected. The VO change manager collaborates with the change initiator to develop the implementation plan which includes all the change details discussed earlier and the initial change schedule. A remediation plan should be developed by the change manager and the change initiator before submitting the request for the change authority authorization. Based on the level of authorization, the change authority will assess the change details and take a decision if the change will be authorized or not.

4.1.5 Coordination and Implementation layer

The VO coordinator will assign a developer to execute and follow the change in conjunction with the release manager. The release manager in the SOVO release management will be responsible to deploy the new changes. A notification will be sent to all VO partners about the status of the change and the change window. This layer consult the SOVO process management framework to facilitate the implementation process of the new changes [16].

4.1.6 Evaluation and Closure layer

The implemented changes need to be evaluated and monitored. This can be done using the SOVO performance management framework's key performance indicators (KPIs) identified in [17] to ensure that risks have been managed and the new modifications meet the requirements. The post implementation review (PIR) document need to be developed which indicate if the change was successful and identifies the opportunities for improvements. The change manager should identify the lessons learned during the process of change which should be stored along with PIR in SOVO knowledge management (KM) repositories.

4.2 Change actors and participants

In order to ensure a successful change management processes and procedures in SOVO, the actors and the participants in the process of change should be identified earlier. We identify the following actors where their roles and responsibilities discussed as part of change control requirements: the change initiator who triggers the RFC, the change manager and change coordinator, the affected partner who are the VO partners that need to be involved in the process of change, finally; the change authority (CA) who authorized the proposed changes.

4.3 Change control requirements

In order to ensure controlled change processes in SOVO six layers of change, some components are needed to be identified, especially in the VO environment. These components include:

4.3.1 Roles and Responsibilities

A list of the roles and responsibilities of each participant in the process of change in the SOVO is essential for successful change management. Furthermore, the responsibility matrix (RM RACI) can be used to identify the Responsible, Consulted, Accountable and Informed participants in the process of change [13].

4.3.2 CAB composition and selection

CAB is a governing body that organized to support in the authorization, assessment, collaboration and decision making of changes in SOVO. The VO coordinator is responsible to compose the CAB members during the creation phase of the VO. The CAB members should be chosen from different partners who are able to deal with both business and technical prospective of change. Each VO business partner has a representative in the CAB, and the VO

change manager is the chair of CAB. The CAB members can trigger changes in the VO environment for different reasons.

4.3.3 Request for change specifications

Identifies the required information that is needed to include in every submitted change request. The RFC specification should be identified and defined by the VO change manager. The ITIL V3 provides a list of all the required parameters which can be customized to be used in initiating a change request in the SOVO.

4.3.4 The level of Authorization

Identifies different levels of authorization linking to the level and impact of a change. The VO change manager along with CAB members will be responsible to identify the level of authorization. It indicates who is responsible to authorize changes in each one of the three levels of change. For example, low level changes do not require the CAB authorization and the change manager may have predefined authority to take action or grant the authorization

4.3.5 Relationship with processes

Identifies ways and methods of tracking and tracing the source of changes in the VO business processes and services by identifying the cause of change, the effect and impact zone of change. If change was triggered from out of the threshold performance indicators, the responsibility zone method can help in identifying the partner responsible for this trigger. This can be done by drilling down to the partner's responsibility zone for this problem using IBM Cognos BI through the SOVO change management console. Furthermore, the impact zone method can be used to identify the severity of change impact on VO operations and goals. The impact zone method determines if change can propagate from a low level of change into a high level of change in the SOVO by identify the depth of influence between each activity and business processes.

4.3.6 Change escalation

It consists of functional and hierarchic escalations used in case a change was taking a long time, difficulty to execute or close. The functional escalation refers to escalating the change problem to people with more skills like an expert. The hierarchical escalation refers to escalating the change problem to higher levels of management like involving the VO manager or the business people of one of the partners in the process of change. The VO coordinator will handle the escalation processes.

4.4 Change related management interfaces

There are different management processes which have primary interaction with the change management process. Some of these management processes are already investigated and implemented in the SOVO environment, which include process management, performance management and knowledge management. Others are still under investigation or prospective research areas. The following briefly describes related management processes.

SOVO Process management: provides a framework to form and manage VOs in a distributed environment based on service oriented architecture. The framework and the implementation infrastructure enable networks of organizations to form virtual workflows based on their software infrastructures, share and monitor their performance metrics without the need for central authority [16]. SOVO change management consults the process management framework during the coordination and implementation layer where the process management framework

provides steps to initiate and manage the VO, and provide solutions which can be used by release management in implementing the new change release.

SOVO Performance management: provides an inter-organizational performance measurement system, which is specifically tailored to the requirements of service oriented virtual organizations[17]. The SOVO change management and the performance management interact with each other's in different stages. Stage one, the performance management provides an input to the change management by triggering a change in one of the levels of change based on the measurements criteria. Stage two, performance management is used to evaluate the implemented changes in the evaluation and closure layer of change management process. Stage three, change management provides the performance management framework with inputs which identified as measurement criteria for the VO collaboration performance. These include the commitment of a partner which represents the willingness of partners to avoid critical situations during the operation of the VO. This includes re-active and pro-active commitment [14]. The flexibility of the VO which describes the early ability to respond to internal or external changes and the ability to adapt to the new situation without escalating the change [14].

SOVO Knowledge management: ensuring that the right information is provided to the right roles at the appropriate time [18]. It ensures that the knowledge developed during the operation and evolution phase of the VO is stored, and can be retrieved when needed. SOVO change management provides an input to knowledge management during the communication and collaboration layer of change process where information and data developed at this stage is captured and stored in the Knowledge management Databases (KMDBs). Furthermore, Post implementation review (PIR) and lessons learned developed in the evaluation and closure layer is stored in the KMDBs. Information and data stored in KMDBs can be used by change management during the identification and assessment layer..

SOVO Problem management: it includes activities to diagnose root causes of events, determine the initial resolution of a problem, request changes (RFC) that resolve those root causes, and reduce the number of future events [18]. Problem management is the trigger of change management processes and procedures.

SOVO Release management: It provides processes to ensure that there are clear and comprehensive deployment plans which bring alignment with customer and business change plans[18]. SOVO change management interacts with release management through the VO coordinator in the coordination and implementation layer to ensure that the proposed changes are installed, tested, and deployed efficiently and on the agreed schedule.

SOVO Configuration management: provides process responsible for managing information about configuration items to deliver a service[18]. SOVO change management interacts with configuration management in the initiation layer to store and recode the configuration item related to change request in the configuration management repositories. Furthermore, change management consults configuration management to assess the impact of a change based on configuration items identified in configuration management processes.

5. IMPLEMENTATION OF THE PROPOSED PROCEDURAL FRAMEWORK

We employ the IBM BPM Process designer to implement change management processes including initiation, identification and assessment, planning and authorization, coordination and implementation, and evaluation and closure as illustrated in Figure 3. We implement the change management process in collaboration and communication layer as collaborative processes between participated partners where integration of communication and collaboration layer collaborative processes with submit-solution-service is done in the IBM BPM Integration designer. Submit-solution-service is a published web service used by partners to submit solutions to the proposed problem. Partners' submitted solutions are imported into the BPM integration designer and used in the change management collaborative processes. As such, the change process is calling (invoke) the submit-solution-service where partners submit their solutions.

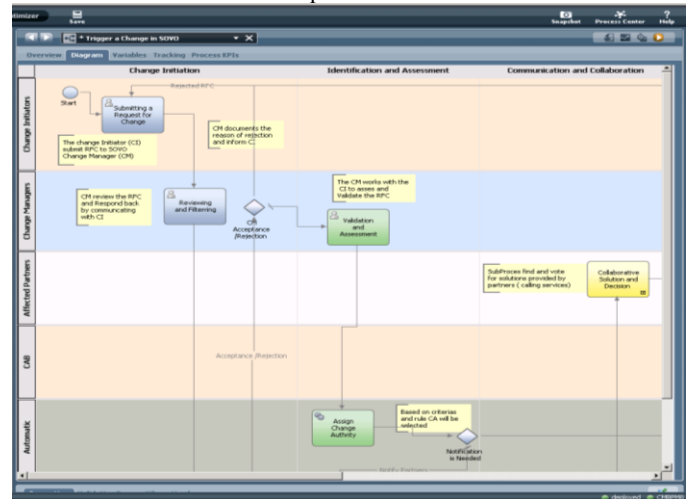


Figure 3: SOVO change processes IBM process designer

Furthermore, we develop a VO Change Management Console using both the IBM BPM business space and process portal. The business space is a browser based Web 2.0 user interface component, which serves as a portal for the change process. Using the business space, the VO CAB members, change manager/coordinator and partners have access to monitor the performance of the VO and the performance of the shared services through the IBM Cognos BI in SOVO based on performance management framework. The process portal allows change process participants to go through the process of change by initiating changes, and supports the process of collaboration and communication. Change Management Console relies on both the business space and the process portal to achieve its goals and objectives. Figure 3 illustrates the change management console architecture used in the prototype implementation.

6. CONCLUSION

The rapidly increasing number of web services, combined with the idea of virtual organizations (VO) to offer new services in a dynamic collaborative environment, introduces new challenges in their change managements. The VO members must constantly monitor the changes in their environment, and react to them in order to meet the market and customers' needs through improved services. With prior knowledge of the impacts and levels of

change, combined with a predefined change processes and procedures which are agreed on between VO participated partners, the VO can respond quickly and flexibly to market demands to remain profitable and competitive.

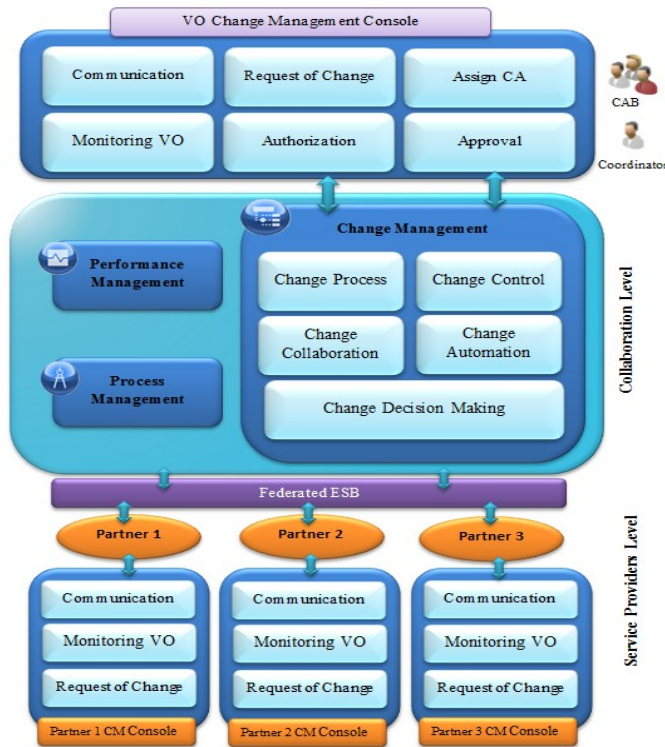


Figure 4: Change management console architecture

We proposed a procedural framework which provided steps by steps, change processes and procedures that VO can follow to initiate changes to their business processes and services in an efficient and predefined way. We presented the implementation of SOVO change processes and change management console architecture. The proposed change management procedural framework, in addition to the structural change management framework can be used together to facilitate the process of change in SOVO to meet its requirement and overcome its challenges. Our next research plan is to investigate scenarios of change or a case study in VO to demonstrate the utility of the proposed solution to solve VO real world change management problems.

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